When a Lesson Does Not Meet the Class: Analyzing Instructional Mismatch Through Individual Learning Theories

I. The Problem of Practice

A lesson in a Hong Kong secondary classroom can be pitched at the wrong level for its students. It may present syllabus content accurately while assuming knowledge the class has not yet developed, or spend too much time on material students already understand. In either case, completing the lesson may produce less learning than intended. This mismatch between instructional demands and students' existing understanding is the problem of practice I examine here.

Several features of Hong Kong's education system may help explain why this mismatch occurs. The first concerns how secondary education is organized. Students are allocated to secondary schools in three bands according to their primary school assessment results, yet every secondary student follows the curriculum leading to the same high-stakes public examination, the Hong Kong Diploma of Secondary Education (Legislative Council Secretariat, 2014). This means that classes with very different starting points are expected to learn content prescribed by the same curriculum. Teachers therefore need to decide how to adapt their instruction to each class when planning their lessons.

Other factors concern the planning process itself. Adapting instruction to a particular class takes time, but heavy workloads and examination pressures often leave teachers with little time to do so. Meanwhile, generative artificial intelligence is changing how teachers plan lessons. A 2026 survey found that more than 20% of Hong Kong's primary and secondary schools were actively exploring the use of AI in teaching, learning, and administration

(Education University of Hong Kong [EdUHK], 2026). When teachers use these tools to draft lesson plans, making those plans suitable for a particular class requires them to provide information about the class or revise the generated content. Research suggests, however, that both steps may receive insufficient attention. In a study of 27 pre-service teachers at a university in eastern China, 59.57% of the information sources in their final lesson plans were AI suggestions adopted without modification (Sun & Huang, 2025). Similarly, when Velander (2026) analyzed the interactions of 60 teachers in Sweden with generative AI, only about one in ten of the teachers' turns gave the tool background about their teaching context. Plans drafted this way may follow the curriculum closely while taking little account of the class.

These conditions help explain how a lesson comes to be planned at the wrong level. What the lesson then does to students' learning is a question for individual learning theories. I draw on cognitive load theory and constructivism to explain why such a lesson can produce little learning and what challenges it creates for students and teachers. I then consider where an account centered on individual learners reaches its limits.

II. How the Theories Illuminate the Problem

Cognitive Load Theory: A Lesson's Difficulty Varies With What a Class Already Knows

Cognitive load theory explains why the same lesson can overwhelm one Hong Kong class and underuse another. Working memory has limited capacity, and de Jong (2010) stresses that intrinsic load, the difficulty that comes from the subject matter, is a function of the prior knowledge a learner brings to the task as well as of the material itself. When a lesson plan is based solely on topics in the curriculum, its demands tend to reflect the prior knowledge that the curriculum assumes students have. Under Hong Kong's banding system for secondary school

allocation, this assumption may hold for some classes but not others. The same lesson plan can therefore impose different levels of cognitive load on different classes.

The theory also explains how mismatch can arise in both directions. When students know less than a lesson assumes, they may have to process too many unfamiliar elements at once, making learning difficult. They still try to make sense of the material. As Rosenshine (2012) explains, misconceptions can develop from these attempts to be logical when background knowledge is weak. The opposite case is harder to notice. As students gain expertise, intrinsic load decreases. Worked examples and step-by-step guidance that help beginners may then become unnecessary for students who already understand the material. Such support can even place an additional burden on more knowledgeable learners. This is known as the expertise reversal effect (Kalyuga et al., 2003, as cited in de Jong, 2010). In this situation, processing unnecessary guidance takes attention away from new material that could extend students' understanding.

On this account, difficulty is a feature of the match between a lesson and a class. The same lesson can be right for one class and wrong for another, and the curriculum alone cannot say which.

Constructivism: Learning Grows From Ideas a Class Already Holds

Constructivism explains mismatch by focusing on how students build understanding from their existing ideas. In Piaget's account, as presented by Siegler (1998), people assimilate new information by adapting it to fit their existing ways of thinking. A lesson therefore needs to connect new content with ideas students already hold. For example, a student may believe that multiplication always makes a number larger. A lesson on multiplying by fractions needs to address this belief so that multiplying eight by one-half makes sense. Otherwise, the student may

learn the procedure without understanding why the result is smaller. Instructional fit therefore depends on how students interpret the content, as well as whether they can carry out the steps.

A lesson pitched too low can create a different problem. In this case, students may be able to understand the material easily, but the lesson gives them little reason to revise or extend their existing thinking. Siegler (1998) notes that equilibration involves movement through disequilibrium toward a more sophisticated mode of thought, and that this process begins when some problem disturbs a child's existing equilibrium. If a lesson stays entirely within what students can already handle, there may be little to create that disturbance. Students can complete the lesson using ideas they already have. A well-fitted lesson therefore needs to be close enough to students' current understanding to make sense, but challenging enough to push that understanding further.

III. What the Theories Reveal About Learning and Teaching Challenges

These explanations also reveal why mismatch is hard to correct, for students and teachers alike.

The Learning Challenge: Mismatch Persists and Accumulates

For students, the first challenge is that they have almost no way of changing a lesson that does not suit them. A student who needs another explanation can ask for help, but that request slows the whole class, and many students stay quiet for that reason. Students may also be unaware of a misconception and have difficulty explaining what they need. Meanwhile, those who understand the material may have few opportunities to move beyond the assigned work. Access to a more suitable task or explanation therefore depends partly on whether the teacher recognizes these different needs. A student can participate and complete the work while still needing an adjustment that neither student nor teacher has identified.

A second challenge is that the effects can accumulate across lessons. When later content builds on earlier learning, missing prerequisite knowledge makes new material harder to process. An unresolved misconception can also shape how the next explanation is understood. Rosenshine (2012) similarly notes that students who have not mastered earlier material may fall further behind as instruction moves on. If these gaps remain unnoticed, later lessons may continue to assume understanding that students are still developing. Conversely, repeated work on already-mastered content can limit opportunities to extend their thinking. Over time, these patterns can leave students with persistent gaps or too few opportunities to progress.

Hong Kong's banding system adds a broader context to this accumulation. Students are allocated to secondary schools by attainment, so a lower-banded school takes in classes that start below what the common syllabus assumes, and a higher-banded school takes in classes that start above it. Each class then meets the same kind of mismatch lesson after lesson, one consistently over-pitched and the other consistently under-pitched. The consequence reaches beyond any single classroom. Where mismatch runs in one direction for a whole cohort, the common curriculum and common examination that were meant to hold all students to one standard end up widening the distance between schools.

The Teaching Challenge: Planning Under Competing Demands

For teachers, the first challenge follows from the students' position. Because students have limited control over a lesson's level, teachers need to anticipate their needs during planning. This is also where a teacher's knowledge of the class can shape the lesson, making it an important opportunity to address mismatch. When AI suggestions enter a plan unmodified, as discussed earlier, teachers still need to check whether the draft reflects what their students know and where they need help.

A second challenge is that the two theories bring different priorities to planning. Cognitive load theory directs teachers to keep demands manageable and adjust support as expertise grows (de Jong, 2010). Constructivism directs teachers to leave room for problems that students' current ideas cannot fully explain, since working through them can help thinking develop. Duckworth (1972) notes that a teacher of thirty or more children can hardly time the right question for each child. She therefore emphasizes settings in which children can raise productive questions themselves. Clear guidance can help students engage with these problems while leaving meaningful thinking for them to do. Judging fit therefore means deciding how much support students need and which challenges they are ready to work through.

The third challenge is time. Estimating what a class knows and preparing a lesson for different starting points both take time, which heavy workloads and examination pressure can limit. A generated plan can save drafting time, but adapting it still requires knowledge of the class and careful review. Velander's (2026) findings suggest that teachers' interactions often focus on requesting content, with fewer exchanges devoted to clarification and refinement. The work of judging whether a lesson fits therefore remains with the teacher, even when producing the draft becomes quicker.

IV. Gaps and Limitations in Applying the Theories to This Problem

Useful as the two theories are for this problem, they leave two questions open.

The first question is how to recognize mismatch in a particular lesson. The theories explain why mismatch hinders learning, but they do not provide a reliable way to identify when it occurs in real classrooms. Signs of struggle might seem an obvious guide, yet research on learning strategies complicates this judgment. Rohrer and Pashler (2010) review strategies such as spacing and interleaving that can lower performance during practice while improving results

on later tests. These effects are described as “desirable difficulties” (Schmidt & Bjork, 1992, as cited in Rohrer & Pashler, 2010). The distinction between immediate performance and later learning suggests that struggle alone is insufficient to identify mismatch. A lesson can also run smoothly while offering little opportunity to extend understanding. My inquiry therefore needs to distinguish difficulty that supports learning from demands students cannot yet manage, using evidence of their understanding over time.

Furthermore, these theories explain learning at the individual level, whereas instructional mismatch occurs within whole classes and schools. The expertise reversal effect suggests that when students in a class differ in prior knowledge, the same level of instructional support may help some while placing an unnecessary burden on others. A lesson can therefore fit a class only approximately, and neither theory tells teachers how to weigh the needs of the students it suits against those it misses. They also leave unexplained why mismatch may concentrate in certain types of schools and how school structures and working conditions shape planning. These questions call for attention to the social and institutional settings of teaching.

V. Conclusion

Instructional mismatch in Hong Kong secondary classrooms is a problem about the relationship between a lesson and the class that receives it. Cognitive load theory shows that the demands of a lesson become manageable in relation to what a class already knows, and constructivism shows that a lesson connects with students where it meets their existing ideas. The syllabus contains neither piece of information. Students can do little to repair such a mismatch during a lesson, and it grows across a term. Teachers must therefore set the level in advance, balancing two competing aims with limited time.

Working through these theories has refined my problem of practice in two ways. I began by treating AI-generated lesson plans as the cause of mismatch. I now see AI-assisted planning as one condition, alongside workload and examination pressure, that shapes how teachers plan and use their knowledge of a class. I also initially associated a well-matched lesson with students being able to follow it easily. The analysis has clarified that a suitable lesson can involve difficulty, provided students have the knowledge and support to work through it. The problem is therefore better stated as lessons planned without reference to what a particular class already understands, which place demands outside the range in which that class can work productively. Whatever produces the first draft, the teacher's knowledge of the class remains central to whether a lesson fits.

References

de Jong, T. (2010). Cognitive load theory, educational research, and instructional design: Some food for thought. *Instructional Science, 38*(2), 105–134. https://doi.org/10.1007/s11251-009-9110-0

Duckworth, E. (1972). The having of wonderful ideas. *Harvard Educational Review, 42*(2), 217–231.

Education University of Hong Kong. (2026, June 23). *EdUHK research reveals positive teacher attitudes on AI, calls for strengthened professional guidance and administrative support* [Press release]. https://www.eduhk.hk/en/press-releases/eduhk-research-reveals-positive-teacher-attitudes-on-ai-calls-for-strengthened-professional-guidance-and-administrative-support

Legislative Council Secretariat. (2014). *Support for students of different learning abilities* (Research Brief Issue No. 7, 2013–2014). Research Office. https://www.legco.gov.hk/research-publications/english/1314rb07-support-for-students-of-different-learning-abilities-20140724-e.pdf

Rohrer, D., & Pashler, H. (2010). Recent research on human learning challenges conventional instructional strategies. *Educational Researcher, 39*(5), 406–412. https://doi.org/10.3102/0013189X10374770

Rosenshine, B. (2012). Principles of instruction: Research-based strategies that all teachers should know. *American Educator, 36*(1), 12–20.

Siegler, R. S. (1998). *Children's thinking* (3rd ed.). Prentice-Hall.

Sun, Y., & Huang, Y. (2025). Exploring the role of generative AI in collaborative lesson planning for pre-service teachers. *Journal of Computer Assisted Learning, 41*(4), e70098. https://doi.org/10.1111/jcal.70098

Velander, J. (2026). Beyond operational skills: Teachers' AI knowledge and interactions with generative AI in lesson planning. *Computers and Education Open, 10*, 100371. https://doi.org/10.1016/j.caeo.2026.100371

AI Use Statement

I used DeepL and ChatGPT for translation, and Grammarly for grammar and proofreading. All ideas and analysis are my own.